AI Engineering Assignment

1. AI Engineering Overview:
Generative AI mainly creates content like text, images, and code. It responds to prompts but does not take actions.
Agentic AI can plan, decide, and take actions using tools and memory.
Example: In e-commerce, Generative AI only replies to customers, but Agentic AI can check orders and process refunds automatically.

2. Environment Setup:
CPU processes tasks one by one and is slower for AI.
GPU can process many tasks at the same time, making AI much faster.
Docker keeps all project environments the same across different machines.

3. LLMs & Quantization:
Quantization makes AI models smaller and faster by reducing memory precision.
It reduces model size and improves speed.
Example: A 14GB model can become 4–6GB after quantization.

4. Prompt Engineering:
System prompts define AI behavior and rules.
Guardrails prevent unsafe or harmful outputs.
They are important for safe and reliable AI systems.

5. RAG Architecture:
RAG retrieves information from external documents before generating answers.
Document processing cleans and organizes data.
Chunking splits documents into small parts for better search and accuracy.